



IoT Based Smart Fan and Lighting System with Temperature Adaptive Control, Voice Assistance and Safety Monitoring

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ABSTRACT

- + • IoT-based smart home system integrating environmental automation and safety monitoring.
- Automatic fan speed control based on room temperature.
- Smart lighting adjustment according to ambient brightness.
- Motion detection system to monitor elderly individuals or children.
- Fall or inactivity detection sends instant alerts to the user's mobile app.
- Hybrid control options: physical regulators, mobile application, and voice commands.
- Improves energy efficiency, modern convenience, and enhances household safety.

INTRODUCTION

- Rapid growth of IoT technology is making household appliances smarter, connected, and autonomous.
- Traditional fans and lighting systems require manual control, leading to inefficient energy usage.
- Smart home automation enables intelligent, sensor-based control of appliances.
- Appliances automatically adjust to environmental conditions like temperature and brightness.
- Incorporates motion detection to monitor the safety of occupants.
- Especially useful for elderly individuals and children who may need continuous monitoring.
- The project focuses on improving energy efficiency and user convenience.

EXISTING SYSTEM

- Traditional switch-based systems where fans and lights are controlled manually.
- Remote-controlled or timer-based home automation devices with limited functionality
- Separate safety devices like panic buttons or wearable trackers for elderly monitoring.
- Mobileapp–controlled smart appliances that allow remote operation.
- Sensor-based lighting systems that automatically turn lights on/off based on motion or brightness.

PROPOSED SYSTEM

- **Temperature Monitoring:** To improve overall energy efficiency through IoT based automation.
- **Adaptive Light Control:** Uses LDR sensors to adapt brightness to the environment, while allowing manual app overrides
- **Safety Monitoring:** Motion sensors detect the presence of individuals and identify abnormal patterns or potential falls.
- **Alert System:** Safety Monitoring: Motion sensors detect the presence of individuals and identify abnormal patterns or potential falls.
- **Voice & Mobile Interface:** Enables remote operation through smartphones and voice-activated commands.

SYSTEM REQUIREMENTS

Hardware Requirement :

- ESP8266 / ESP32 Microcontroller
- Temperature Sensor (DHT11 / DHT22)
- PIR Motion Sensor
- LDR (Light Dependent Resistor) / Light Sensor
- Relay Module
- Fan
- LED / Light Bulb (for brightness control demonstration)
- Manual Regulator / Potentiometer
- Breadboard and Connecting Wires
- Power Supply (5V Adapter)

SYSTEM REQUIREMENTS

SOFTWARE REQUIREMENT:

- Programming Language:
- Embedded C / Arduino Programming Development Tools:
- Arduino IDE
- Blynk / MIT App Inventor for mobile application
- IFTTT (for Google Assistant integration) Operating System:
- Windows

TECHNOLOGIES USED

- **Communication** for mobile app connectivity.
- **Mobile Application** for remote monitoring and manual control of devices.
- **Cloud or Alert System** to send notificationIoT (**Internet of Things**) for connecting devices and enabling smart control.
- **Temperature and Light Sensors** to automatically adjust fan speed and light intensity.
- **Motion/Gesture Sensors** to detect movement or emergency gestures.
- **Microcontroller (Arduino / NodeMCU / ESP8266)** for processing sensor data and controlling devices.
- **Wi-Fi / Bluetooth** s during emergency situations.

EXPECTED OUTCOME

- Automatic fan speed adjustment based on temperature.
- Remote fan control through mobile application.
- Voice control of the fan using Google Assistant.
- Manual fan control using a physical regulator.
- Motion detection for monitoring elderly people and children.
- Alert notifications for unusual inactivity.
- Automatic light intensity/brightness control based on surrounding light conditions.
- Improved energy efficiency and smart home automation.

REFERENCES

- [1].Gurmu M. Debele and Xiao Qian,"Automatic Room Temperature Control System Using Arduino UNO R3 and DHT11Sensor," School of Mechanical and Electrical Engineering, University of Electronic Science and Technology of China, 2020.
- [2].Sree Suvro Kumar Biswas, Arpita Sarkar Singdha, Labanya Talukder, Md. Hafizur Rahman, Prangon Das, and Md. Manirul Islam, "Sustainable Energy-Based Voice Controlled Home Automation Using IOT," Department of Mechatronics Engineering, Rajshahi University of Engineering and Technology, Bangladesh.
- [3].Dr. S. Kavitha, Dr. J. V.Archithaalagammai, S. Murali, P. Shreemathi, and L. Yazhinipriya, "IoT-Based Centralized System for Street Light Automation and Fault Detection," Velammal College of Engineering and Technology, Tamil Nadu, India.
- [4].Sneha Shelke, Dipesh Bhaurao Pardeshi, Sarthak Wake, and Rahul Rajendra Bibve, "ThermoBreezePro: Cost Effective Remote Speed Controlled Fan with Intelligent Temperature Sensing for Seamless Comfort," Department of Electrical Engineering, Sanjivani College of Engineering, Maharashtra, India.
- [5] .Dilruba Yamin, "Automatic Room Temperature Controlled Fan Speed Controller Using PT100," International Journal of Scientific and Engineering Research, Vol. 6, Issue 8, 2015.

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